Adding and Subtracting in Scientific Notation

Question Bank - Additional 4.01

Name_____
Date_____
Period_____

Show your work. Circle your final answer.

For questions 1 through 6, add or subtract. Write each answer in scientific notation.

1. $(5.3 \times 10^{7}) + (2.4 \times 10^{7})$

2. $(9.6 \times 10^{-3}) - (4.2 \times 10^{-3})$

3. $(7.4 \times 10^{5}) + (8.9 \times 10^{5})$

4. $(6.3 \times 10^{-7}) - (5.9 \times 10^{-7})$

5. $(3.8 \times 10^{9}) + (6.1 \times 10^{8})$

6. $(2.7 \times 10^{-2}) - (9.0 \times 10^{-4})$

Real quantities.

7. Mars's diameter is about 6.79×10^{3} kilometers. Earth's is about 1.28×10^{4} kilometers.

a. What is the sum of the two diameters?

b. How much larger is Earth's diameter than Mars's?

8. A charity raised about 4.85×10^{5} dollars in its first year and about 1.27×10^{6} dollars in its second.

a. How much did it raise in the two years together?

b. How much more did it raise in the second year than in the first?

Working backwards.

9. Find the missing number in each equation.

a. $(5.2 \times 10^8) + (n \times 10^8) = 1.34 \times 10^9$

b. $(9.6 \times 10^{-4}) - (n \times 10^{-4}) = 3.8 \times 10^{-4}$

c. $(6.5 \times 10^3) + (3.5 \times 10^3) =$ _____

Reading someone else's work.

10. Malik writes $(3.2 \times 10^5) + (7.1 \times 10^7) = 1.03 \times 10^{13}$. Name the two mistakes he made, and give the correct sum.

11. Which is the value of $(6.8 \times 10^{-6}) + (5.4 \times 10^{-6})$, written in scientific notation?

- ○ A. 1.22×10^{-5}
- ○ B. 12.2×10^{-12}
- ○ C. 12.2×10^{-6}
- ○ D. 1.22×10^{-12}

12. Select ALL of the expressions whose value is 2.4×10^6 .

- ☐ A. $(1.2 \times 10^6) + (1.2 \times 10^6)$
- ☐ B. $(1.8 \times 10^6) + (6 \times 10^6)$
- ☐ C. $(2.42 \times 10^6) - (2 \times 10^4)$
- ☐ D. $(1.8 \times 10^6) + (6 \times 10^5)$
- ☐ E. $(2.5 \times 10^6) - (1 \times 10^5)$
- ☐ F. $(2.5 \times 10^6) - (1 \times 10^6)$